

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards

(BL2,BL5)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

*I **N. HEMA SREE (192511100)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: N. HEMA SREE

Annexure A

Constraint-Based Problem Solving

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability

ANSWER:

Introduction

A multinational company is establishing a branch office in a remote rural area where high-speed wired Internet is unavailable. The network must support cloud applications, video conferencing and secure communication with headquarters while maintaining low cost, reliability and future scalability.

Requirements and Constraints

- Wired broadband is unavailable.
- Cloud applications require continuous Internet connectivity.
- Video conferencing requires stable bandwidth and low delay.
- Communication with headquarters must be secure.
- The budget is limited.
- Business operations should continue even if the primary connection fails.
- The network should support future expansion.

Proposed Network Design

I recommend using **4G/5G Fixed Wireless Access** as the **primary Internet connection**. If affordable, a **satellite connection** can be used as a **backup**.

Network Diagram

CLOUD / INTERNET

|

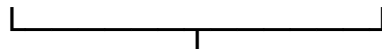
4G / 5G Router

|



| Dual-WAN |

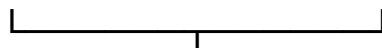
| Router |



|



| Firewall |

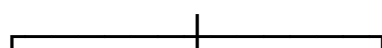
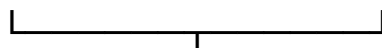


|



| Managed |

| Switch |



| | |

PCs APs Printers

|

IPsec VPN

|

Headquarters

Recommended Devices

1. **4G/5G Router** – Provides Internet where wired broadband is unavailable.
2. **Dual-WAN Router** – Provides automatic switching between primary and backup connections.
3. **Next-Generation Firewall** – Protects the branch from unauthorized traffic and attacks.
4. **Managed Switch** – Connects computers, printers and access points and supports VLANs.
5. **Wi-Fi 6 Access Point** – Provides wireless access to employees.

Security Mechanisms

1. IPsec VPN

A secure VPN tunnel should be established between the branch and headquarters. It encrypts confidential business data travelling through the Internet.

2. Firewall

The firewall should provide:

- Access control
- Stateful inspection
- Intrusion prevention
- VPN support
- Web/application filtering

3. VLANs

Different users and devices should be separated.

VLAN 10 → Employees

VLAN 20 → Management

VLAN 30 → Voice/Video

VLAN 40 → Guest Network

Guest users should not be allowed to access company resources.

4. QoS

Quality of Service (QoS) should give higher priority to video conferencing and cloud applications.

Video Conference → Highest Priority

Cloud Applications → High Priority

Email/Web → Medium Priority

Downloads/Updates → Low Priority

Justification

Constraint	Solution
No wired Internet	4G/5G
Limited budget	Wireless primary connection
Connection failure	Dual-WAN/backup connection
Video conferencing	QoS
Secure HQ communication	IPsec VPN
Unauthorized access	Firewall
Guest isolation	VLAN

Constraint

Solution

Future growth

Scalable router and managed switch

Conclusion

The proposed **4G/5G + Dual-WAN Router + Firewall + Managed Switch + Wi-Fi 6 + IPsec VPN + VLAN + QoS** solution provides a good balance of **bandwidth, reliability, security, cost and scalability**. It is suitable for a rural branch office where wired Internet is unavailable.

2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.

ANSWER:

Introduction

A university requires campus-wide wireless Internet access for more than 3,000 students in classrooms, laboratories, libraries and hostels. The network must support many simultaneous users while reducing interference and maintaining secure access with a limited number of access points.

Requirements and Constraints

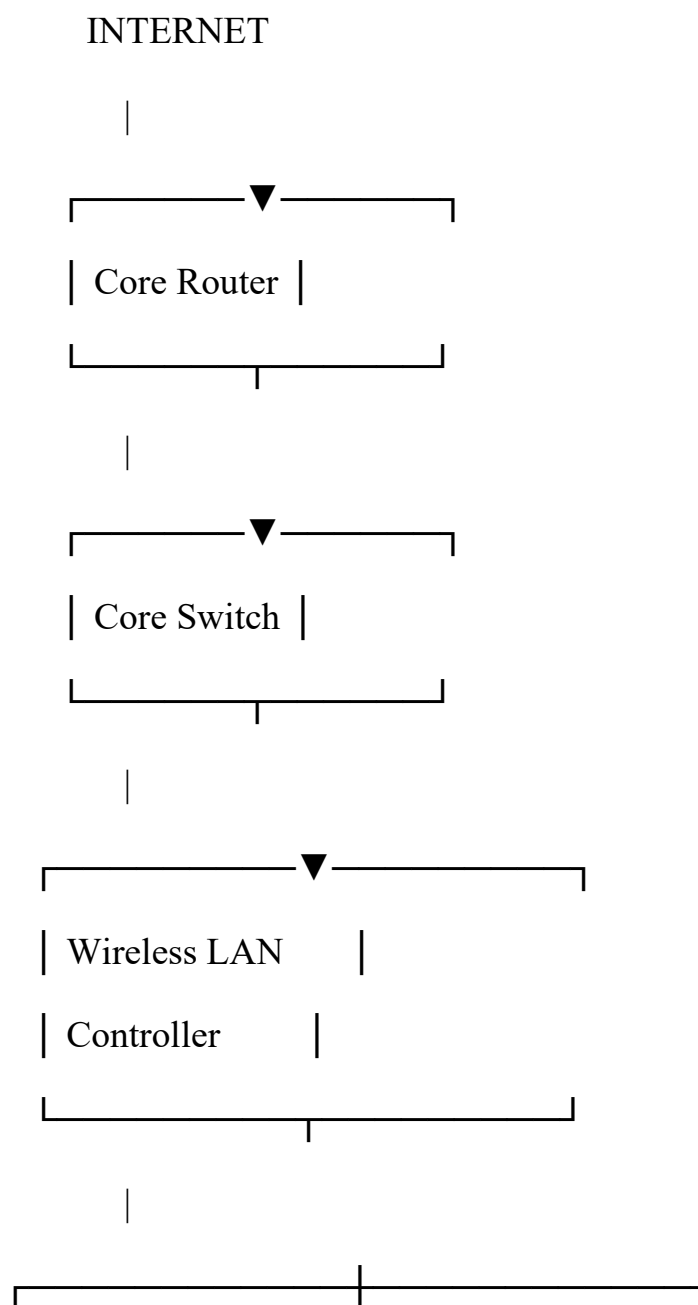
- More than 3,000 users.
- Campus-wide coverage.
- High number of simultaneous connections.
- Limited number of access points due to budget.
- Minimum signal interference.

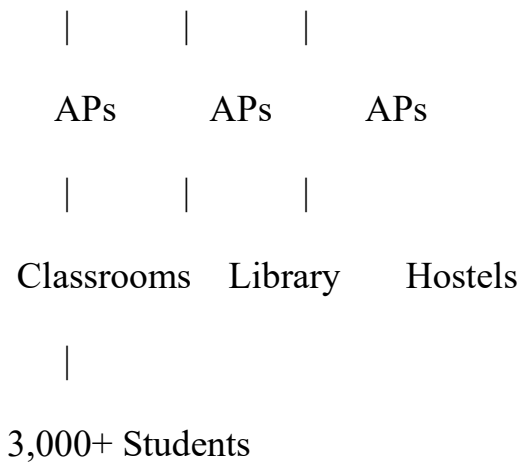
- Secure authentication.
- Good performance and future scalability.

Proposed Network Design

A **centrally managed Wi-Fi 6/6E network** should be deployed. Access points should be placed based on **user density rather than simply geographical coverage**.

Network Diagram





Access Point Placement

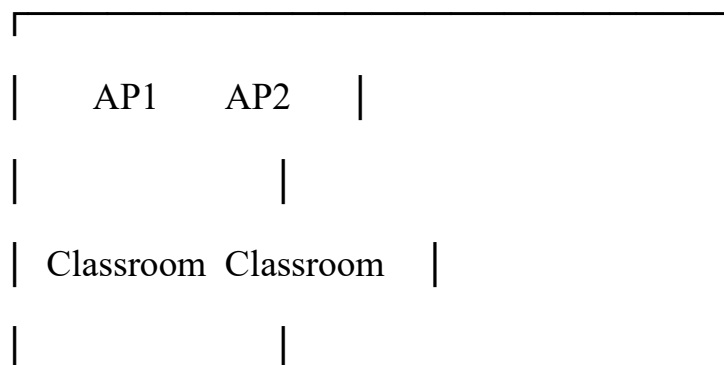
APs should be concentrated in high-density locations such as:

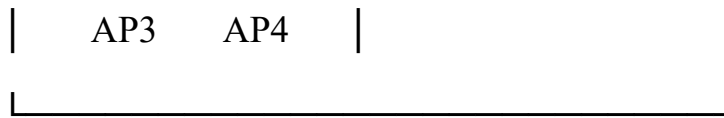
- Lecture halls
- Laboratories
- Libraries
- Hostels
- Student common areas

Fewer APs can be used in low-density areas such as corridors and administrative spaces.

Example

CLASSROOM BLOCK





A wireless site survey should be performed before final installation.

Frequency Band Selection

5 GHz

5 GHz should be the primary band because it provides:

- Higher speed.
- More available channels.
- Less interference than 2.4 GHz.
- Better performance for high-density areas.

2.4 GHz

2.4 GHz can be retained for older devices and devices requiring greater range.

6 GHz

Where supported, 6 GHz can be used for newer devices and high-density areas.

Channel Planning

For 2.4 GHz, the commonly used non-overlapping channels are:

Channel 1	Channel 6	Channel 11
AP1	AP2	AP3

For 5 GHz/6 GHz, suitable non-overlapping channels should be assigned to nearby APs.

Transmit power should also be controlled to reduce interference.

Authentication and Security

The university should use:

WPA3-Enterprise + 802.1X + RADIUS authentication

Student Device

|

↓

Access Point

|

↓

RADIUS Server

|

↓

University Identity Database

This provides individual authentication instead of using one common Wi-Fi password.

VLAN Segmentation

VLAN 10 → Students

VLAN 20 → Faculty

VLAN 30 → Administration

VLAN 40 → Guest Users

VLAN 50 → IoT Devices

This prevents students and guests from directly accessing sensitive university systems.

Performance Optimization

To efficiently use the limited APs:

- Use Wi-Fi 6/6E high-capacity APs.
- Use band steering.
- Use load balancing.
- Use airtime fairness.
- Control transmit power.
- Use proper channel allocation.

Justification

Constraint	Solution
3,000+ students	High-capacity Wi-Fi 6 APs
Limited APs	Density-based placement
Interference	Channel planning
High performance	5 GHz/6 GHz
Security	WPA3-Enterprise
Authentication	802.1X + RADIUS
User isolation	VLANs
Future scalability	Wi-Fi 6E/7-ready equipment

Conclusion

A centrally managed Wi-Fi 6/6E network with density-based AP placement, 5 GHz/6 GHz frequency selection, channel planning, WPA3-Enterprise, 802.1X, RADIUS and VLAN segmentation provides an effective balance of coverage, performance, security and cost.

3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

ANSWER:

Introduction

A financial institution handles highly sensitive customer and banking information. Therefore, the network must provide strong security while maintaining good performance, meeting compliance requirements and controlling operational costs.

Requirements and Constraints

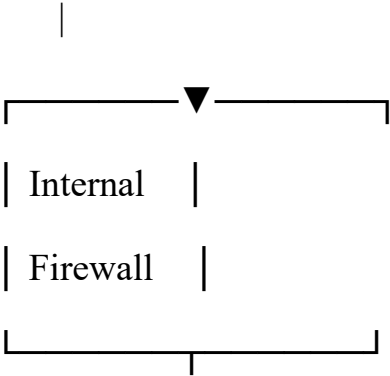
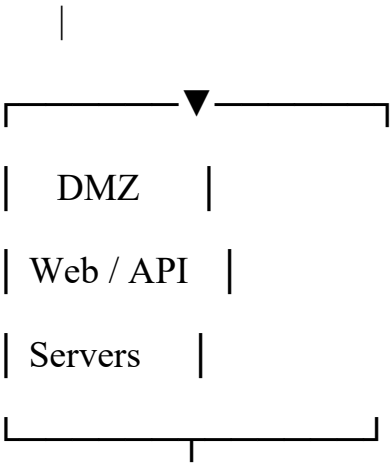
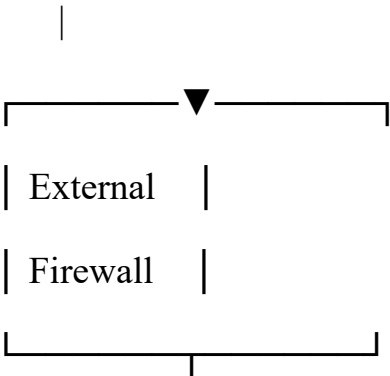
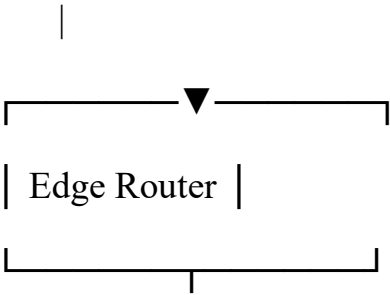
- Protect sensitive customer information.
- Secure online banking services.
- Allow authorized employees to access required information.
- Prevent unauthorized access.
- Detect network attacks.
- Meet security and compliance requirements.
- Avoid unnecessary performance degradation.
- Control security costs.

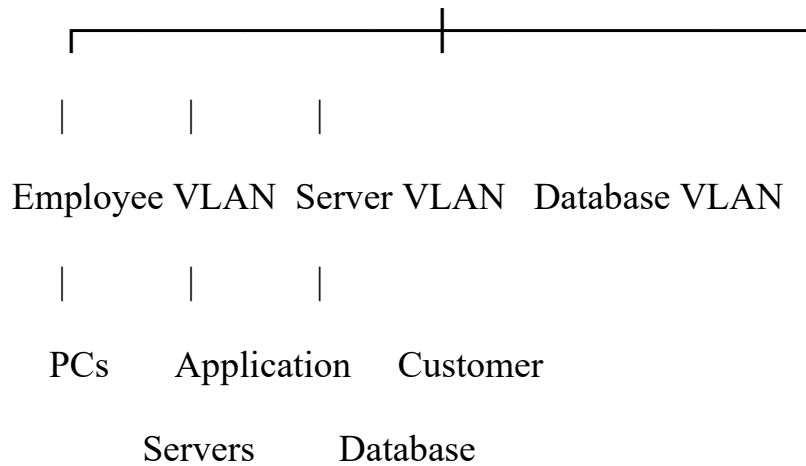
Proposed Secure Network Architecture

A **defense-in-depth architecture** using VLANs, DMZ, firewalls, MFA, IDS/IPS and access control should be implemented.

Network Diagram

INTERNET





1. Network Segmentation

VLANs should separate different departments and critical resources.

VLAN 10 → Employees

VLAN 20 → Finance

VLAN 30 → Application Servers

VLAN 40 → Database Servers

VLAN 50 → Network Management

VLAN 60 → Guest Network

Access between VLANs should be controlled using firewall rules and ACLs.

For example:

Employee PC



Employee VLAN



Firewall / ACL



Application Server



Database Server

Employees should not directly access the database unless explicitly authorized.

2. Firewall Deployment

Two levels of firewall protection should be used.

External Firewall

Protects the organization from Internet-based attacks.

Internal Firewall

Controls traffic between internal departments, application servers and databases.

DMZ

Public-facing web and API servers should be placed in a **DMZ**.

The customer database must never be directly exposed to the Internet.

3. Authentication

Multi-Factor Authentication

Employees accessing sensitive applications should use:

Username + Password

+

OTP / Authenticator

+

Device Verification

802.1X

802.1X should be used for secure wired and wireless network access.

Role-Based Access Control

Users should receive only the permissions required for their job.

Teller → Transaction System

Manager → Reports + Approval

IT Admin → Network Administration

Guest → Internet Only

This follows the **principle of least privilege**.

4. IDS/IPS

An **Intrusion Detection and Prevention System** should monitor network traffic.

Internet



Firewall



IDS / IPS



DMZ



Internal Firewall



Internal Network

It can identify activities such as:

- Port scanning.
- Brute-force attacks.
- Malware traffic.
- Suspicious login attempts.
- Abnormal data transfers.

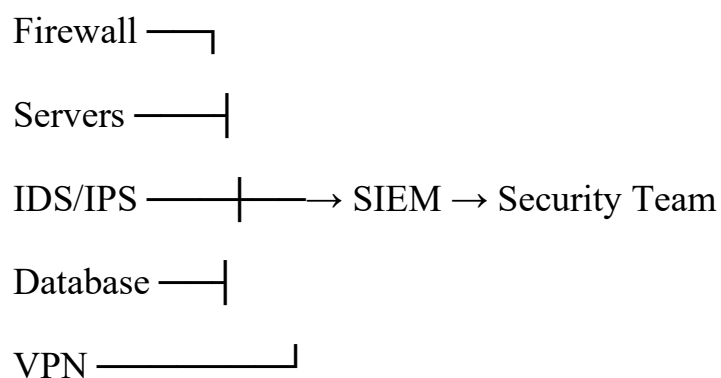
5. Encryption

Sensitive data should be protected using:

- HTTPS/TLS for web communication.
- IPsec VPN for remote connections.
- Secure protocols for administration.
- Encryption for sensitive data at rest.

6. Centralized Monitoring

Firewall, server, database and IDS/IPS logs should be collected centrally.



This improves monitoring, auditing and incident response.

Application of Constraints

Constraint

Solution

Sensitive customer data Segmentation + encryption

Constraint	Solution
Internet attacks	Firewall
Public banking services	DMZ
Unauthorized access	MFA + 802.1X
Excessive privileges	RBAC
Internal threats	VLANs + ACLs
Attack detection	IDS/IPS
Compliance	Logging and auditing
Performance	Efficient firewall and network segmentation
Budget	Centralized security management

Security, Performance and Cost Balance

The design avoids placing every security function everywhere. Critical systems such as **database servers and banking applications receive the strongest protection**, while ordinary traffic is handled efficiently.

Existing managed switches can be used for VLAN segmentation, reducing additional hardware costs. Centralized monitoring also reduces administration effort.

Conclusion

The recommended architecture consists of:

VLAN Segmentation + DMZ + External/Internal Firewalls + MFA + 802.1X + RBAC + IDS/IPS + Encryption + Centralized Logging/SIEM.

This provides a strong **defense-in-depth security architecture** while balancing **security, performance, compliance and budget**.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Problem & Constraints	20%	Clearly identifies all constraints and fully understands the problem requirements	Identifies most constraints with minor gaps in understanding	Identifies some constraints but misses key aspects	Shows little or no understanding of constraints or problem
Problem Decomposition	15%	Breaks problem into logical, well-structured sub-problems effectively	Breaks problem into sub-parts with some logical flow	Attempts decomposition but lacks clarity or structure	No clear decomposition of the problem
Application of Constraints in Solution	20%	Applies all constraints correctly and consistently in solution design	Applies most constraints correctly with minor errors	Applies some constraints but with noticeable errors	Fails to apply constraints appropriately
Logical Reasoning & Strategy	15%	Uses strong, clear, and efficient reasoning strategies	Uses generally sound reasoning with minor gaps	Reasoning is partially correct but inconsistent	Reasoning is unclear or incorrect
Solution Accuracy & Feasibility	15%	Solution is fully correct, feasible, and optimized	Solution is mostly correct and feasible	Solution has errors or is partially feasible	Solution is incorrect or not feasible
Creativity & Innovation	5%	Demonstrates highly innovative and unique approach	Shows some creativity in approach	Limited creativity shown	No creativity; relies on basic or copied ideas

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Clarity of Presentation	10%	Solution is clearly presented with excellent organization and explanation	Presentation is clear with minor issues	Presentation lacks clarity or organization	Poorly presented and difficult to understand